

Annex C

Epistemic Convolutional Neural Networks (ECNN) Formalisation in a Reticular Local Abstraction Perspective

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Abstract

This annex presents a formal specification of *Epistemic Convolutional Neural Networks* (ECNN). The aim is to extend classical convolutional networks with an explicit epistemic channel that can represent explanation, undecidability and contradiction, while remaining fully Turing-computable and compatible with the *Compact Reticular Computability* (CRC) framework introduced in Annex A. The construction is grounded in a multi-level lattice view: convolutional and pooling layers are interpreted as levels of a *Reticular Local Abstraction* (RLA), and their non-injective maps are treated as controlled coarse-grainings. On top of the representation channel, an *Epistemic Computational Unit* (ECU) and an epistemic head implement rule-based reasoning: they can accept or reject the network’s own predictions and explicitly say “unknown” when the evidence is insufficient. In more advanced configurations, ECUs are realised as orchestrated assemblies of Large Language Models (LLMs) acting as high-level “epistemic neurons”, each specialised on a deterministic epistemic matrix. The annex includes: (i) definitions and axioms; (ii) bridging theorems linking CNNs and ECNNs; (iii) an architectural template with ECU and epistemic head; (iv) training protocols; (v) metrics for performance, emergence and falsifiability, including layer-wise collapse and emergence indices compatible with those of Annex A; (vi) a roadmap for minimal case studies. Throughout, ECNNs are understood as *epistemic theories*: their internal structures discretise scientific knowledge about a domain, without claiming to describe its ultimate ontology.

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1 Introduction and Motivation

Convolutional neural networks (CNNs) have become a standard tool in machine learning, particularly for vision and sequential data (LeCun, Bottou, Bengio, & Haffner, 1998; Goodfellow, Bengio, & Courville, 2016). In their textbook form, CNNs form a linear chain of layers: convolutions, pooling, fully connected blocks and an output head. This pipeline emphasises predictive accuracy and feature compression, but it does not explicitly represent logical contradiction, undecidability, or explanations in a falsifiable way.

At the same time, multi-level descriptions of complex systems in physics, biology and cognitive science suggest that emergent properties may depend crucially on non-injective mappings between levels (Anderson, 1972; Noble, 2012; Godfrey-Smith, 2007). Pooling operations in CNNs, when viewed through this lens, are not merely engineering tricks: they are coarse-grainings that may create macro-features not transparently reducible to single microstates.

Epistemic Convolutional Neural Networks (ECNNs) are proposed as a way to integrate these insights within the RLA/CRC framework of Annex A. Starting from a classical CNN, an ECNN:

- reinterprets convolution and pooling layers as levels of a Reticular Local Abstraction (RLA);
- introduces an *epistemic layer* that can encode explanation, contradiction and “I do not know”;
- in advanced forms, realises high-level neurons as LLM sessions constrained by deterministic epistemic matrices;
- remains fully Turing-computable and internally falsifiable, yielding a compact reticulum in the sense of Compact Reticular Computability (CRC).

The purpose of this annex is to provide a compact and self-contained formalisation of ECNNs suitable as a technical reference and as a bridge between classical deep learning and the reticular viewpoint.

2 Background: CNNs and Reticular Local Abstraction

2.1 Classical CNN notation

Let $\mathcal{X}$ be the input space (e.g. images or token sequences) and $\mathcal{Y} = \{1, \dots, K\}$ a finite set of labels. A dataset is a finite multiset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \subset \mathcal{X} \times \mathcal{Y}.$$

A classical CNN defines a function $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^K$ with parameters θ , built as a composition of layers

$$f_\theta = h_L \circ h_{L-1} \circ \dots \circ h_1,$$

where h_ℓ are convolutions, pooling layers, or fully connected maps. The final softmax layer yields a categorical distribution $p_\theta(y \mid x)$, trained via cross-entropy.

2.2 Reticular Local Abstraction: minimal recall

For ECNNs we adopt a minimal lattice formalism that is consistent with Annex A.

Definition 2.1 (Reticular Local Abstraction). *A Reticular Local Abstraction (RLA) is a finite family of levels $L = \{L_0, \dots, L_n\}$ with associated state spaces $D(L_i)$, together with transmission functions*

$$f_{i \rightarrow j}: D(L_i) \longrightarrow \mathcal{P}(D(L_j)),$$

where $\mathcal{P}$ is the powerset. We write $L_i \preceq L_j$ if there exists a finite path $L_i \rightarrow \dots \rightarrow L_j$ formed by such transmissions.

A map $f_{i \rightarrow j}$ may be injective, quasi-injective, or non-injective over relevant states. Non-injectivity is interpreted as a controlled loss of micro-level information, enabling emergent macro-properties.

Definition 2.2 (Compact lattice). *An RLA is compact (in the sense of CRC) if:*

1. *all explanatory and predictive work is internal to its levels (Internal Ontological Autonomy, IOA);*
2. *any observational statement about the model's domain can be expressed as a predicate over some $D(L_i)$ (Epistemic Closure, EC);*
3. *all transmissions and update rules are Turing-computable (Turing-Computability, TC).*

This annex assumes RLA only at this minimal level; no additional machinery from Annex A is required beyond IOA, EC and TC.

3 Definition of Epistemic CNN (ECNN)

3.1 Layered structure

Definition 3.1 (Epistemic CNN). *An Epistemic Convolutional Neural Network (ECNN) is a tuple*

$$\mathcal{E} = (L_0, \dots, L_n, L_{\text{ep}}, \Sigma)$$

where:

1. $L_0, \dots, L_n$ are levels corresponding to standard CNN layers: input, convolution, pooling, dense blocks, etc.;
2. L_{ep} is at least one additional level for the epistemic channel;
3. Σ is a finite set of Turing-computable update rules and transmissions.

For each $i < n$ there is a transmission $f_{i \rightarrow i+1}: D(L_i) \rightarrow \mathcal{P}(D(L_{i+1}))$ representing the action of layer L_{i+1} on states of L_i . There is also an epistemic map $\varepsilon: D(L_n) \rightarrow D(L_{\text{ep}})$.

Intuitively, L_0 encodes raw data, intermediate levels $L_1, \dots, L_{n-1}$ implement feature extraction and compression, L_n provides high-level logits or representations, and L_{ep} carries epistemic states (explanations, contradictions, “unknown”).

3.2 ECNNs as CRC instances

Proposition 3.1 (ECNNs as compact reticula). *Let $\mathcal{E} = (L_0, \dots, L_n, L_{\text{ep}}, \Sigma)$ be an ECNN with:*

- *a fixed epistemic matrix M encoded in a finite representation;*
- *a representation channel B_θ implemented by finitely many convolutional, pooling and dense layers;*
- *an ECU and epistemic head defined by deterministic algorithms (including any LLM calls with fixed prompts, temperature, and parsing rules).*

If the resulting system satisfies IOA and EC with respect to a chosen epistemic horizon (set of observables and questions), then $\mathcal{E}$ is a compact RLA reticulum and hence an engineered instance of Compact Reticular Computability (CRC) in the sense of Annex A.

Sketch. Turing-computability (TC) follows because each component of the ECNN representation channel and epistemic head is implemented by a finite algorithm operating on finitely representable reals or finite strings. Internal Ontological Autonomy (IOA) holds if all explanatory entities and laws for the domain are represented by levels L_i and transmissions in Σ , including the epistemic matrix M . Epistemic Closure (EC) holds if every observable and epistemic query is encoded as a predicate over states in $D(L_i)$ or $D(L_{\text{ep}})$. Under these conditions, the tuple of levels and transmissions satisfies the definition of a compact reticulum given in Annex A, and CRC applies. $\square$

When $(L_0, \dots, L_n, L_{\text{ep}}, \Sigma)$ is used under such IOA/EC assumptions, we speak of a *compact ECNN reticulum*.

3.3 Convolution and pooling as lattice maps

Definition 3.2 (Convolutional level). *A convolutional level L_c has a state space $D(L_c) = \mathbb{R}^{H \times W \times C}$ and a set of kernels $\{K_1, \dots, K_M\}$. The associated transmission $f_{c \rightarrow c+1}$ is defined by*

$$f_{c \rightarrow c+1}(x) = \{\text{Conv}(x)\}, \quad \text{Conv}(x)(p) = \sum_{\delta} K(\delta) x(p + \delta),$$

where the map is quasi-injective up to a symmetry group G (e.g. translations, reflections) on a discriminative subset of states.

Definition 3.3 (Pooling level). *A pooling level L_p has state space $D(L_p)$ identical in shape to its input. The transmission $f_{p \rightarrow p+1}$ is*

$$f_{p \rightarrow p+1}(x) = \{\text{Pool}(x)\}, \quad \text{Pool}(x)(p) = \text{agg}\{x(q) : q \in \mathcal{N}(p)\},$$

where agg is max or average over a neighbourhood $\mathcal{N}(p)$. Over relevant states, $f_{p \rightarrow p+1}$ is non-injective.

Non-injectivity induced by pooling is the main mechanism through which micro-configurations can collapse into macro-features.

3.4 Epistemic layer

Definition 3.4 (Epistemic layer). *An ECNN possesses at least one epistemic level L_{ep} with state space $D(L_{\text{ep}})$ and a map $\varepsilon: D(L_n) \rightarrow D(L_{\text{ep}})$. Within $D(L_{\text{ep}})$ there is a distinguished subset of states $\Xi \subseteq D(L_{\text{ep}})$ corresponding to:*

- *epistemic labels (e.g. accept, reject, unknown);*
- *structured explanations or rationales;*
- *explicit contradiction markers.*

The epistemic layer is the minimal locus where the network can express its own ignorance or detect conflict between internal states.

3.5 Coherence and coverage

Theorem 3.1 (Classical CNN as a special case). *If the levels $L_0, \dots, L_n$ form a linear chain without an epistemic layer and all transmissions are single-valued, then the ECNN reduces to a classical CNN. In this case, non-injectivity is restricted to pooling and there is no meta-epistemic channel.*

Sketch. Without L_{ep} and with deterministic $f_{i \rightarrow i+1}$, the tuple $(L_0, \dots, L_n, \Sigma)$ induces a standard function $f_\theta: \mathcal{X} \rightarrow \mathbb{R}^K$ by composing all layers. This coincides with a conventional CNN in the sense of LeCun et al. (1998) and Goodfellow et al. (2016). $\square$

4 Minimal Formalism: Axioms and Bridging Theorems

4.1 Axioms

We state four axioms capturing the intended behaviour of ECNNs as multi-level computational systems. They specialise the general axioms of Annex A to the case of neural architectures.

Axiom 4.1 (Computable levels). *For each level L_i there exists a Turing machine that computes $f_{i \rightarrow i+1}$ on any finite representation of states in $D(L_i)$. This ensures the absence of non-computable oracles in the network and implements TC.*

Axiom 4.2 (Propagation of undecidability). *If some subset of $D(L_i)$ encodes a Turing-undecidable problem, and if $f_{i \rightarrow i+1}$ is quasi-injective over that subset, then undecidability is preserved at L_{i+1} . A downstream level cannot solve an undecidable problem without violating computability.*

Axiom 4.3 (Non-injectivity and emergence). *If $f_{i \rightarrow i+1}$ is non-injective over a set of relevant microstates, then there exists at least one macro-property at L_{i+1} that does not correspond to a single microstate at L_i . Such a property is called emergent relative to L_i .*

Axiom 4.4 (Epistemic closure). *In a compact ECNN, every observable phenomenon and every interaction with the external world is mediated by some level in the lattice or by the epistemic channel. No explanation or contradiction requires appeal to extra-lattice entities.*

Philosophically, these axioms connect computability theory (Church, 1936; Turing, 1936), emergentism (Anderson, 1972; Godfrey-Smith, 2007) and a Popperian view of falsifiability (Popper, 1959).

4.2 Bridging theorems

Theorem 4.1 (Convolution and pooling as lattice maps). *In any ECNN satisfying Axioms 1 and 3:*

- *each convolutional level realises a quasi-injective transmission that preserves undecidability where present;*
- *each pooling level realises a non-injective quotient map, implementing the conditions of Axiom 3 and enabling emergent macro-features.*

Sketch. Convolutions are local linear operators; when restricted to relevant patches and up to symmetries, they do not identify distinct inputs, so they are quasi-injective. Typical pooling operators, by contrast, map many micro-configurations to the same macrostate, thus satisfying the non-injectivity condition. $\square$

Theorem 4.2 (Epistemic layer and falsifiability). *Assume Epistemic Closure. If an ECNN includes a non-trivial epistemic layer L_{ep} and a map $\varepsilon: D(L_n) \rightarrow D(L_{\text{ep}})$, then the network can internally:*

1. *flag contradictions induced by external challenge sets;*
2. *represent undecidable situations via explicit “unknown” states.*

Sketch. Given any input x , the state at L_n is mapped by ε to a state in $D(L_{\text{ep}})$. Epistemic Closure ensures that every relevant conflict can be expressed at that level. Thus, whenever a challenge set triggers incompatible requirements (e.g. a pair of images that should not share a label), the epistemic layer can encode a contradiction or an undecidable state instead of a forced label. $\square$

5 Architecture of ECNNs in Practice

5.1 Representation channel

Let the *representation channel* be a standard CNN

$$B_\theta: \mathcal{X} \longrightarrow \mathbb{R}^d,$$

obtained by stacking convolutions, pooling operations and possibly dense layers. The vector $z = B_\theta(x)$ represents a high-level embedding of the input.

5.2 Epistemic matrix

The epistemic behaviour is governed by an *epistemic matrix* M capturing the domain theory.

Definition 5.1 (Epistemic matrix). *An epistemic matrix is a tuple*

$$M = (E, R, C, P),$$

where:

- E is a finite set of epistemic entities (concepts, predicates, labels);
- R is a finite set of inference rules between entities;
- C is a finite set of coherence constraints (incompatibilities, subsumptions, etc.);
- P is a set of policies for rejection, abstention and default behaviour.

All elements of M are given in a representation that can be processed by a Turing machine.

M determines which predictions are allowed, which are contradictory, and when “unknown” is the appropriate answer.

5.3 Epistemic Computational Unit (ECU)

Definition 5.2 (Epistemic Computational Unit). *Given a representation $z = B_\theta(x)$ and an epistemic matrix M , an Epistemic Computational Unit (ECU) is a deterministic function*

$$\text{ECU}_\phi: \mathbb{R}^d \times M \longrightarrow \mathcal{A},$$

where $\mathcal{A}$ is a finite space of epistemic artifacts. Each artifact $a \in \mathcal{A}$ consists of:

- a primary decision $y^* \in \mathcal{Y} \cup \{\perp, \perp_c\}$, where $\perp$ denotes “unknown” and $\perp_c$ “contradiction”;
- an optional confidence score $q \in [0, 1]$;
- a structured explanation or rationale, encoded in a finite symbolic format.

The ECU interprets z as a set of evidential claims, checks them against M , and decides whether the evidence suffices for a label, suggests ignorance, or detects a contradiction.

5.4 Epistemic head

The final *epistemic head* H_ψ maps artifacts into output distributions:

$$H_\psi: \mathcal{A} \rightarrow \Delta(\mathcal{Y} \cup \{\perp, \perp_c\}),$$

where $\Delta(S)$ denotes the set of probability distributions on S . The overall ECNN is then

$$F_{\theta, \phi, \psi}(x) = H_\psi(\text{ECU}_\phi(B_\theta(x), M)).$$

5.5 LLM-based epistemic neurons

In advanced configurations, the ECU is implemented as a network of LLM-based “epistemic neurons”.

Definition 5.3 (Epistemic neuron). *An epistemic neuron is an LLM instance with parameters ω , called with:*

1. a standardised prompt template,

2. a fixed temperature (typically $T = 0$),
3. a request to emit a JSON object with fields constrained by M .

The raw text output is parsed by a deterministic validator that enforces syntactic and semantic coherence with the matrix M . The resulting structured object is a contribution to the artifact a .

A finite set of such neurons, each specialised on a subdomain of M , can be orchestrated by a deterministic controller (itself an ECU component) to produce global artifacts. Although individual LLMs are powerful black boxes, the overall ECU remains Turing-computable when the inference protocol (prompts, temperature, parsing) is fixed.

Conceptually, this implements the intuition that “neurons” at high levels of an ECNN can themselves be epistemic agents, not just scalar units. The emergent behaviour depends on the *topology* of the epistemic network, not on the sheer number of low-level parameters.

5.6 Practical deterministic regime for LLM-based ECUs

In principle, large language models are often deployed in stochastic modes (e.g. with nonzero temperature or sampling variability). For the purposes of Compact Reticular Computability, however, ECNNs assume a *practical deterministic regime* for any LLM-based ECU:

- the underlying LLM checkpoint (architecture and parameters) is fixed and versioned;
- decoding parameters such as temperature, top- k and top- p are set to values that make the model effectively deterministic (e.g. $T = 0$, fixed k and p), and any random seed is held constant for a given configuration;
- prompt templates, system instructions and few-shot examples used by epistemic neurons are specified, versioned and treated as part of the epistemic matrix M ;
- parsing and validation of LLM outputs (e.g. JSON schemas, type checkers, consistency checks against M) are implemented by deterministic procedures.

Under these assumptions, each LLM-based epistemic neuron behaves as a deterministic transducer from finite input strings (features, prompts, matrix M) to finite structured artifacts. Real-world deployments may deviate from this idealisation (for instance because of hardware nondeterminism or changing model checkpoints), but the conceptual model of ECNNs treats ECUs as Turing-compatible components. Stochasticity or drift in practice is then regarded as part of the external environment, not as an intrinsic violation of the CRC assumptions.

6 Training Protocols

6.1 Datasets and labels

Training data are extended to include epistemic supervision. Each datum has the form

$$(x_i, y_i, e_i),$$

where $y_i \in \mathcal{Y}$ is a standard label and $e_i \in \{\text{certain}, \text{uncertain}, \text{contradictory}\}$ is a meta-label. Optionally, a structured target explanation can be provided for a subset of points.

Additional *challenge sets* Ω are defined to test falsifiability. We write:

- Ω_{CP} for the set of *contradictory pairs*: pairs (x, x') such that assigning the same label would violate constraints in M ;
- Ω_{UT} for the set of *undecidability triggers*: inputs engineered so that no consistent label in $\mathcal{Y}$ is justifiable from M ;
- Ω_{SA} for *semantic adversaries*: perturbations that preserve low-level statistics but invert or scramble semantic content.

6.2 Loss functions

A generic training objective combines standard classification, epistemic penalties and explanation regularisation:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{unk}} \mathcal{L}_{\text{unk}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}} + \lambda_{\text{exp}} \mathcal{L}_{\text{exp}}.$$

Classification loss. On examples marked as “certain”, $\mathcal{L}_{\text{cls}}$ is standard cross-entropy:

$$\mathcal{L}_{\text{cls}} = - \sum_{i \in I_{\text{cert}}} \log F_{\theta, \phi, \psi}(x_i)_{y_i},$$

where I_{cert} is the set of indices with $e_i = \text{certain}$.

Unknown penalty. On “uncertain” examples, the network is rewarded for abstaining:

$$\mathcal{L}_{\text{unk}} = - \sum_{i \in I_{\text{unc}}} \log F_{\theta, \phi, \psi}(x_i)_{\perp},$$

where I_{unc} is the set of indices with $e_i = \text{uncertain}$.

Contradiction penalty. On contradictory pairs (x, x') in Ω_{CP} the loss penalises unflagged contradictions:

$$\mathcal{L}_{\text{con}} = \sum_{(x, x') \in \Omega_{\text{CP}}} \mathbf{1}[\text{ECNN does not output } \perp_c] \cdot \alpha,$$

for some constant $\alpha > 0$. Flagging a contradiction (via $\perp_c$) is treated as correct behaviour.

Explanation regularisation. Where ground-truth explanations E_i are available, a penalty encourages artifacts a_i to match them (e.g. via token-level cross-entropy or symbolic distance). Otherwise, a complexity term discourages gratuitous verbosity.

7 Metrics of Performance, Emergence and Falsifiability

7.1 Standard performance metrics

Besides accuracy, precision and recall, calibration measures such as expected calibration error (ECE) assess the reliability of probabilities from $F_{\theta, \phi, \psi}$.

7.2 Emergence metrics

To track emergent behaviour we introduce:

- **Feature compression ratio:** the ratio of input dimension to the dimension of $z = B_{\theta}(x)$, combined with information-theoretic measures when available.
- **Representation collapse index:** the average pairwise cosine similarity between embeddings of distinct classes; high values may indicate loss of discriminative microstructure.
- **Gradient collapse instability:** the mean norm of gradients at critical convolution and pooling layers across training; persistent growth resembles the amplification of self-organised macroscopic patterns (Carpenter et al., 2011).

These indicators mirror early-warning signals for regime shifts in complex systems (Scheffer et al., 2009).

7.3 Layer-wise collapse and emergence indices

To connect ECNNs quantitatively with the collapse and emergence diagnostics of Annex A, we define layer-wise adaptations of the *coefficient of collapse* and the *index of emergence*.

Let L_i and L_j be two consecutive levels in the representation channel (e.g. a pooling layer followed by a convolutional or dense layer), with transmission $f_{i \rightarrow j}$. Let $S \subset D(L_i)$ be a finite reference set of states, sampled from activations of a validation set.

Coefficient of collapse. The empirical coefficient of collapse from L_i to L_j is

$$CC_{i \rightarrow j}^{(\text{ECNN})}(S) := 1 - \frac{|f_{i \rightarrow j}(S)|}{|S|} \in [0, 1],$$

where $f_{i \rightarrow j}(S)$ is the set of distinct outputs observed in $D(L_j)$ for inputs in S (up to a chosen numerical tolerance). A value near zero indicates near-injective behaviour on S , while a value close to one indicates strong activation collapse. This is a layer-wise analogue of the coefficient defined in Annex A.

Index of emergence. Let $\mathcal{Q}$ be a finite family of candidate macro-properties (e.g. cluster assignments, linear probe classifications, or logical predicates over embeddings) defined on $D(L_j)$. Each property $Q \in \mathcal{Q}$ can be tested for definability as a predicate on $D(L_i)$, using the expressive resources of the models available at L_i . The empirical index of emergence is

$$IE_{i \rightarrow j}^{(\text{ECNN})} := \frac{\#\{Q \in \mathcal{Q} : Q \text{ is computable at } L_j \text{ and not definable on } D(L_i)\}}{|\mathcal{Q}|} \in [0, 1].$$

A high value indicates that many behaviourally relevant properties only appear clearly at the higher layer, in line with the notion of emergent macro-features in Annex A.

Reticular irreducibility. At the network scale, one can construct a causal graph G_t over layers and units (or feature groups) and estimate a reticular irreducibility index RIR over a training horizon T in analogy with Annex A, by computing normalised minimum vertex cuts that separate input and output units across time. While detailed definitions are left to Annex D, the key idea is that large RIR signals a structurally irreducible lattice of computations across layers, compatible with Wolfram-style computational irreducibility.

These indices make it possible to compare ECNNs with other CRC reticula using a common diagnostic language.

7.4 Falsifiability metrics

Falsifiability requires that the ECNN can fail in observable ways when confronted with challenge sets.

- **Contradiction rate (CR).** On a set of contradictory pairs Ω_{CP} ,

$$CR = \frac{\#\{\text{unflagged contradictions}\}}{\#\Omega_{CP}}.$$

Lower values indicate better detection of contradictions.

- **Undecidability miss rate.** On undecidability triggers Ω_{UT} , the fraction of cases where the network returns a definite label instead of $\perp$.
- **Epistemic coverage.** The proportion of real-world inputs for which the ECNN emits a determinate label (not $\perp$) while remaining within the bounds of M .
- **Explanation coherence.** Using logical or statistical checks, the fraction of explanations that satisfy the coherence constraints C in M .

Together, these metrics quantify how well the ECNN operationalises Popperian falsifiability internally.

7.5 Relation to uncertainty and abstention methods

ECNNs do not replace existing approaches to uncertainty quantification and abstention; rather, they provide a reticular architecture and an explicit epistemic channel in which such methods can be embedded. Table 1 summarises the relation.

In this sense, ECNNs are best understood as a structural generalisation of existing uncertainty-aware classifiers: the representation channel behaves like a conventional CNN (possibly Bayesian or ensemble-based), while the epistemic channel L_{ep} hosts explicit rules, policies and challenge-set responses that turn uncertainty summaries into falsifiable epistemic states $(y, \perp, \perp_c)$ with explanations.

Method family	Integration in ECNN epistemic channel L_{ep}
Bayesian / deep ensembles	Posterior variances, ensemble disagreement scores or predictive entropy are computed from B_{θ} (or from an ensemble of representation channels) and passed as inputs to the ECU. The epistemic matrix M can include rules such as “abstain if ensemble disagreement $>$ threshold”, and the epistemic head H_{ψ} maps these signals to $\perp$ or to down-weighted label distributions.
Selective classification / reject option	Classical reject-option schemes (e.g. Chow’s rule) provide thresholds on confidence or risk. In an ECNN, these thresholds become policies P inside M and are implemented explicitly in L_{ep} : the ECU checks whether the current evidence satisfies the accept region; if not, it outputs $\perp$ (unknown) or $\perp_c$ (contradiction) instead of forcing a label.
Conformal prediction	Conformal p-values or prediction sets are computed from the base representation channel and treated as epistemic signals. The ECU incorporates them into its artifact (e.g. by annotating labels with calibrated coverage guarantees), and the epistemic head enforces that only labels supported by valid conformal sets are asserted, while the others trigger abstention or contradiction markers.

Table 1: Schematic relation between standard uncertainty / abstention methods and the ECNN epistemic channel L_{ep} . ECNNs provide a reticular structure and an explicit locus for epistemic decisions, rather than a new “miraculous” uncertainty algorithm.

8 Minimal Case Studies and Roadmap

8.1 Minimal application domains

The formalism is intended to be domain-agnostic. Minimal testbeds include:

- **Image classification with abstention:** an ECNN trained on a restricted class set with challenge images outside its domain;
- **Textual entailment with contradictions:** inputs are sentence pairs; labels are *entails*, *contradicts*, *unknown*; LLM-based neurons specialise in linguistic phenomena;
- **Toy scientific theories:** small symbolic models (e.g. cellular automata or toy ecology) where rules in M are exactly known, allowing precise tests of emergence and undecidability.

8.2 Roadmap

Future directions suggested by the formalism include:

1. **Multi-agent lattices:** several ECNN modules exchange epistemic artifacts and negotiate contradictions, forming higher-order lattices.
2. **Meta-learning of contradictions:** training ECNNs not only to avoid inconsistencies but to learn typical “shapes” of paradoxical inputs and recognise them.

3. **Neuromorphic optimisation:** mapping ECNN components to hardware that supports explicit abstention and contradiction states, reducing latency and power.

9 Philosophical Conclusions

ECNNs treat undecidability, emergence and ignorance as first-class computational citizens. Rather than hiding paradoxes in opaque numerical failures, the epistemic channel forces the network to *admit* when its knowledge is inadequate or internally inconsistent. In this sense:

- they realise Anderson’s insight that “more is different” (Anderson, 1972) within a precise computational framework;
- they resonate with Noble’s idea of “no privileged level of causation” (Noble, 2012), since emergent properties arise from non-injective mappings across levels;
- they embed Popperian falsifiability in the network’s own structure, rather than treating it as an external philosophical requirement.

When ECUs are built from LLM-based epistemic neurons, the resulting systems acquire a layered form of meta-reasoning: each neuron is itself a bounded reasoner obeying the matrix M , and the overall topology determines what kinds of epistemic states can emerge. This points toward a view of artificial cognition in which *topology*, rather than raw scale, becomes the key variable for understanding complex behaviour.

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